

Network Security Fundamentals

1. Network Security Fundamentals

Network security consists of the policies and practices adopted to prevent and monitor unauthorized access, misuse, modification, or denial of a computer network and network-accessible resources.

2. Types of Attacks

- DoS (Denial of Service): Overloading a system's resources so it cannot respond to legitimate service requests.
- MITM (Man-in-the-Middle): An attacker secretly relays and possibly alters the communications between two parties who believe they are directly communicating with each other.
- Phishing: Fraudulent attempts to obtain sensitive information by disguising as a trustworthy entity in electronic communication.

3. Firewalls and VPNs

A firewall is a network security device that monitors and filters incoming and outgoing network traffic based on an organization's previously established security policies.

A VPN (Virtual Private Network) extends a private network across a public network and enables users to send and receive data across shared or public networks as if their computing devices were directly connected to the private network.

4. Encryption and SSL/TLS

- Symmetric Encryption: The same key is used to both encrypt and decrypt the information.
- Asymmetric Encryption: Uses a public key for encryption and a private key for decryption.

SSL/TLS (Secure Sockets Layer/Transport Layer Security) are cryptographic protocols designed to provide communications security over a computer network, widely used in HTTPS.

5. Authentication and Best Practices

Authentication verifies the identity of a user, device, or system. Common methods include passwords, biometrics, and Multi-Factor Authentication (MFA).

Best practices include regular security audits, keeping systems patched and updated, using strong passwords, network segmentation, and employee training on security awareness.